Original Manuscript ID: NAIST_Thesis-2024-0208

Original Article Title: "Improving Home Appliance-Based Electricity Theft Detection: Insights from Real and Synthetic Attack Scenarios"

To: Thesis Committee

Re: Response to reviewers

Dear Thesis Committee,

Thank you for allowing a review and resubmission of my dissertation manuscript, with an opportunity to address the reviewers' comments.

I will be uploading (a) our point-by-point response to the comments (below) (response to reviewers, under "Author's Response Files*"*), (b) an updated manuscript with yellow highlighting indicating changes (as "Highlighted PDF*"*), and (c) a clean updated manuscript without highlights ("Main Manuscript"*).*

Best regards,

Abraham Olufemi.

Reviewer#1- Prof Yuichi Hayashi sensei, Concern # 1: Attack scenarios limited at the distribution board? Explain!

- **Author response:** In the context of a Synthetic Binary Discriminator Model (SYNBDM) used for detecting electricity theft and ensuring appliance usage authentication in smart homes, various attack scenarios can threaten the integrity, availability, and confidentiality of the system. These threats are not limited to physical interactions with the distribution board but can also involve digital intrusions and manipulative tactics. Here are some potential attack scenarios

- **Physical Attacks:**
  1. Meter Swapping: Replacement of meters with units from unoccupied or low-usage locations.
  2. Power Diversion: Redirecting power supply within a neighborhood.
  3. Meter Tampering: Includes various methods such as meter removal/disconnection, reversing the meter's position, using magnets to affect readings, and unauthorized access to Smart Meters (SMs).
- **Cyber Attacks:**
  1. Credential Theft: Unauthorized access to meters using stolen login credentials.
  2. Firmware Hacking: Remotely hacking into Smart Meter firmware.
  3. Data Tampering: Altering meter storage data including total energy consumption, audit logs, and encryption keys.
- **Data Attacks:**
  1. Zero/Negative Reporting: Falsely reporting zero or negative energy consumption.
  2. Consumption Report Alteration: Stopping or altering energy usage reports.
  3. Measurement Exclusion: Removing high-consumption appliances from measurements.

Author action: We updated the manuscript by addressing classification of attacks in conventional and smart grids systems in section 2.4, page 23, subsection 4.3.1, page 56, and section 5.3, pages 72-73.

Reviewer#1-Prof Yuichi Hayashi sensei, Concern # 2: How do you address privacy concerns in your proposed model?

Author response: To address privacy concerns with the use of aggregated power consumption patterns for training a Synthetic Binary Discriminator Model (SYNBDM), it is essential to emphasize the process of aggregation and anonymization to ensure individual data cannot be reverse-engineered. Implementing robust security measures and ensuring compliance with data protection regulations are crucial steps in protecting user privacy. Engaging transparently with stakeholders and continuously monitoring and updating privacy practices in response to feedback and new developments will help build trust and ensure ethical use of data.

Author action: We updated the manuscript by inserting subsection 5.4.2-Overview of the feature selection, at page 77, and sections 5.9 privacy concerns and model development at pages 112 – 114.

Reviewer#1-Yuichi Hayashi sensei, Concern # 3: How do you authenticate some appliances that has physical fingerprint with our model?

Author response: Some appliances have unique, measurable characteristics in their electricity usage that can be observed without internal data access, akin to a "fingerprint." In training the SYNBDM, these physical fingerprints—patterns in power consumption—are crucial. They serve as the dataset's features, enabling the model to learn and distinguish between normal operation and anomalies, such as electricity theft. This approach leverages the appliance's consumption behavior as a distinctive signature to improve detection and authentication processes within the smart home environment.

To combine appliance physical fingerprints with consumption patterns in training the SYNBDM model, you would first collect detailed usage data for each appliance, including power draw, usage cycles, and any unique electrical signatures. This data forms the physical fingerprint. Next, integrate this with consumption patterns—how and when the appliance is typically used. The model is trained on these combined features to recognize normal operating conditions. By learning the nuanced differences between appliances and their typical usage, the model can more accurately detect anomalies indicative of electricity theft, enhancing its predictive capabilities.

Author action: We updated the manuscript by adding authentication methods in section 5:10 at pages 113 - 116.

Reviewer#1-Prof Yuichi Hayashi sensei, Concern # 4: How does your model updates?

Author response: Our model update procedure is included in section 5:12.

Author action: We updated the manuscript by including implementation of SYNBDM algorithm page 119 - 122.

Reviewer#1-Prof Yuichi Hayashi sensei, Concern # 5: Did you use another dataset to train your model?

Author response: In evaluating our synthetic attack model, we trained our model with UTokyo Data - Real Attack Data. We consider including the consumption pattern of real rooms of the University of Tokyo as attack data for the model tests. We measured the power consumption at the power distribution boards of the I-REF building (6th-floor building) from 2012 with a sampling frequency of 1 min as described in section 5.5.6 at page 90. Table 5.2 shows the distribution of simulated and real binary datasets at same page.

Author action: We updated the manuscript by including UTokyo Data - Real Attack Data in subsection 5.6.6 at page 91.

Reviewer#2-Prof Keiichi Yasumoto, Concern # 1: Does your model extend the attacks level beyond smart home to AMI?

Author response: Implementation of our ETD model in Advance Metering Infrastructure (AMI) was explained in section 5.13.

Author action: We updated the manuscript by adding visualization of anonymized dataset, Figures 5.19, 5.20, 5.21, 5.22 and 5.23 at pages 123-127.

Reviewer#2-Prof Keiichi Yasumoto, Concern # 2: Suggestion to visit Yamana Lab, Waseda University

Author response: Four papers were extracted from Professor Yamana laboratory namely: 1) Privacy-Preserving Data Falsification Detection in Smart Grids using Elliptic Curve Cryptography and Homomorphic Encryption, 2) Fully Homomorphic Encryption with Table Lookup for Privacy-Preserving Smart Grid, 3) Look-Up Table based FHE System for Privacy Preserving Anomaly Detection in Smart Grids, 4) Towards Privacy-

Preserving Anomaly-based Attack Detection against Data Falsification in Smart Grid. The comparison of the methodologies used in these papers as related to our model is shown in the table below:

Paper Name	Methods Used	Contributions	Limitations	Practical Implications
Electricity Theft Detection for Smart Homes: Harnessing the Power of Machine Learning with Real and Synthetic Attacks	SYNBDM-XGB, RF, MLP, LUM-MLP-AE, 1D-CONV-AE, Isolation Forest	High accuracy in theft detection with AUC scores up to 98.74%. Expands model to detect unknown attacks.	Dependence on data quality, overfitting, privacy concerns.	Enhances energy security and management in smart homes.
ElectricityTheftDetectioninAMIUsing	SVM, CPBETD algorithm, Kernel function, Cluster analysis	Novel algorithm for detecting energy theft in AMI. Robust against contamination attacks.	False alarm rate, need for on-site inspection, privacy risks with high sampling rate.	High-performance solution for energy theft detection in smart grids.
Fully_Homomorphic_Encryption_with_Table_Lookup_for_Privacy-Preserving_Smart_Grid	FHE, Table Lookup with FHE, Integer encoding	Efficient protocol for function evaluation with FHE. More practical than previous methods.	Limited to addition and multiplication on encrypted data. Lookup table scalability.	Protects user data in smart grids while allowing function evaluation.
Privacy-Preserving_Data_Falsification_Detection_in_Smart_Grids_using_Elliptic_Curve_Cryptography_and_Homomorphic_Encryption	ECC-based homomorphic encryption	18x faster execution than CKKS scheme. Ensures data privacy with smaller memory space.	Long execution time for HE schemes. Privacy concerns with customer data analysis.	Fast and privacy-preserving detection of data falsification in smart grids.
Look-Up_Table_based_FHE_System_for_Privacy_Preserving_Anomaly_Detection_in_Smart_Grids	Homomorphic LUT-based FHE, Private information retrieval with FHE	First implementation of flexible control over detection accuracy and time. Detects various attack types.	Differential privacy limits accuracy. SMC has high communication costs.	Enables privacy-preserving anomaly detection with flexible accuracy and time control.
Towards Privacy-preserving Anomaly-based Attack	Optimization of encryption for resource-constrained devices, Homomorphic encryption	Framework detects energy theft and data integrity attacks. 40x faster encryption for low-power devices.	Balancing privacy and security. Future work on quantification of information leakage.	Optimized for automated billing and load monitoring in smart grids, preserving user privacy.

Author action: We updated the manuscript by this summary which encapsulates the core aspects of each paper, making it easier to compare their methodologies, contributions, limitations, and the practical implications of their findings, especially in relation to the first paper – our model in subsection 5.11.12 at page 118.

Reviewer#3-Associate Prof Yuzo Taenaka sensei, Concern # 1: The original concerns of consumers is to conserve money in the pocket. Discuss the different between prevention and consumption of the system

Author response: In the context of electricity theft detection in a smart home, "consumption" and "prevention" have specific meanings related to the management and safeguarding of electrical resources:

1. **Consumption**: This refers to the usage of electricity by various devices and systems within the smart home. Consumption patterns can be monitored and analyzed for various purposes, including efficiency optimization, cost calculation, and detecting anomalies. Smart meters and home energy management systems can track which devices are using electricity, how much they are using, and at what times. This data is crucial for understanding normal electricity usage patterns in the home.

2. **Prevention (in the context of electricity theft)**: This involves measures and strategies to prevent unauthorized use or theft of electricity. Electricity theft can occur in various ways, such as physically tampering with meters, bypassing meters, or hacking into smart metering systems. Prevention strategies include:

    - **Technical Measures**: These could be improved security features in smart meters, such as tamper detection sensors, encryption of data transmission, or secure authentication protocols.

    - **Data Analysis and Anomaly Detection**: By analyzing consumption data, unusual patterns that might indicate theft can be detected. For example, a significant unexplained increase in consumption or patterns that don't match the expected usage profile of the household might trigger an investigation.

    - **Regular Inspections and Audits**: Physical inspections of meters and wiring can help detect tampering or bypass attempts.

Author action: We updated the manuscript by including different between prevention and consumption of the system in section 5:15 at page 134.

Reviewer#3-Associate Prof Yuzo Taenaka sensei, Concern # 2: What is the consumption pilot approach?

Author response: In this context, the relationship between consumption and prevention is critical. Analyzing electricity consumption data not only helps in managing and optimizing energy use but also plays a vital role in the prevention of electricity theft. By understanding normal consumption patterns, anomalies that might suggest theft can be more easily identified and investigated.

The term "consumption pilot approach" in the context of electricity usage, particularly in smart homes or smart grids, typically refers to a pilot study or a trial program aimed at understanding, monitoring, and optimizing electricity consumption. This kind of approach is often used to test new technologies, methodologies, or behavioral interventions before a broader rollout.

Author action: We updated the manuscript by including the consumption approach in subsection 5.15.1 at page 135.

Reviewer#3-Associate Prof Yuzo Taenaka sensei, Concern # 3: How can you compare your model with other scenarios?

Author response: Our model performance comparison was highlighted in section 5.14, pages 128 - 133

Author action: We updated the manuscript by adding Tables 5.9, 5.10, and 5.11. Figures 5.24, 5.25, and 5.26.

Reviewer#4-Prof Youki Kadobayashi sensei, Concern # 1: Dissertation heavily compiled; section numbers and acknowledgment missing.

Author response: All missing sections and acknowledgement has been dully addressed

Author action: We updated the manuscript by correcting the missing section and adding acknowledgement at page 146 – 147.

Reviewer#4-Prof Youki Kadobayashi sensei, Concern # 2: Dissertation not accepted yet pending the approval of journal for publication.

Author response: Our submitted manuscript to IEEE Access journal has been accepted for publication

Author action: We updated the manuscript by adding journal Accepted to the list of paper publications at page 158.

Reviewer#4-Prof Youki Kadobayashi sensei, Concern # 3: Attack model lacking (the model demonstrated is safe in Japan)

Author response: The attack model has been discussed in section 2.4, page 23 and how model could mitigate against these threat models for physical, cyber and data attacks as discussed in our model-SYNBDM implementation Algorithm 1 in section 5.12.1.

Author action: We updated the manuscript by adding algorithm for attack model at page 122.

Reviewer#4-Prof Youki Kadobayashi sensei, Concern # 4: **What is the economic implication of your model?**

Author response: The economic implication of SYNBDM deployment was discussed in section 5.15.

Author action: We updated the manuscript by adding economic implications at pages 135-137.
